

ANNISTON, AL, USA

WMO: 722287

Lat: 33.587N

Lon: 85.856W

Elev: 181

StdP: 99.17

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 13871

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.2	-4.0	-14.2	1.1	-1.8	-11.8	1.4	0.6	8.8	9.8	8.0	9.5	2.0	350	0.377

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.6	34.9	23.7	33.7	23.7	32.6	23.6	25.8	31.4	25.3	30.8	24.8	30.0	2.9	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.2	19.5	28.1	23.8	19.0	27.6	23.3	18.4	27.0	80.2	31.5	77.9	30.8	75.9	30.1	29.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.5	6.4	5.6	-9.9	36.4	2.3	1.6	-11.6	37.5	-12.9	38.4	-14.3	39.3	-15.9	40.5		
(5)			-10.8	26.8	2.3	0.9	-12.4	27.5	-13.8	28.0	-15.0	28.4	-16.7	29.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	17.3	6.8	9.0	12.9	17.1	21.5	25.2	26.7	26.6	23.7	17.7	11.8	8.3		
	DBStd	8.18	5.70	5.27	4.93	3.90	3.27	2.11	1.66	1.94	3.06	4.38	4.64	5.20		
	HDD10.0	372	134	76	28	2	0	0	0	0	0	2	33	97		
	HDD18.3	1457	359	262	176	67	13	0	0	0	3	65	200	311		
	CDD10.0	3044	34	48	119	216	356	456	518	513	410	242	88	45		
	CDD18.3	1087	1	2	8	31	109	206	260	255	163	46	5	1		
	CDH23.3	9835	1	8	86	316	972	1846	2438	2395	1372	371	28	3		
	CDH26.7	4060	0	0	10	57	319	797	1119	1114	556	86	3	0		
(14) Wind	WSAvg	2.3	2.7	2.8	2.8	2.6	2.1	1.9	1.8	1.7	2.0	2.1	2.3	2.6		
(15) Precipitation	PrecAvg	1325	128	119	154	128	114	108	113	93	89	68	102	111		
	PrecMax	1776	322	228	460	439	292	236	310	323	201	160	262	262		
	PrecMin	709	24	30	35	4	46	0	20	13	8	0	24	18		
	PrecStd	234	55	56	93	80	57	62	53	61	53	43	50	54		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.7	24.6	28.0	30.0	32.8	35.2	36.3	37.2	35.3	30.8	26.4	23.5		
		MCWB	16.6	17.5	18.3	19.9	22.5	23.0	24.1	24.5	22.0	20.3	18.2	18.4		
	2%	DB	20.6	22.1	25.7	28.3	31.2	33.7	34.5	35.0	33.4	28.8	23.8	21.3		
		MCWB	15.9	16.3	17.1	18.8	21.8	23.3	24.3	24.3	22.1	20.3	17.1	17.2		
	5%	DB	18.4	20.2	23.8	26.9	30.0	32.5	33.2	33.5	31.8	27.3	22.1	19.4		
		MCWB	15.0	15.1	16.0	18.2	21.3	23.3	24.4	24.0	22.2	19.1	16.6	15.8		
	10%	DB	16.3	18.0	21.7	25.2	28.6	31.2	32.1	32.2	30.0	25.4	20.3	17.2		
		MCWB	13.1	13.4	15.1	17.4	20.8	23.1	24.2	23.7	22.0	18.5	15.7	14.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.7	19.1	20.1	22.1	24.5	25.8	26.8	26.7	24.7	23.5	21.3	19.9		
		MCDB	21.4	22.8	24.1	27.3	29.8	31.2	32.3	33.4	29.8	27.7	23.1	22.4		
	2%	WB	17.1	17.9	18.8	20.8	23.5	25.1	25.8	25.8	24.1	22.3	19.5	18.2		
		MCDB	19.3	21.0	23.3	25.3	28.7	30.7	31.3	31.7	29.1	26.2	22.0	20.4		
	5%	WB	15.5	16.6	17.7	19.9	22.6	24.6	25.4	25.2	23.6	21.4	18.1	16.6		
		MCDB	17.9	19.3	21.9	24.4	27.7	30.0	30.8	30.7	28.3	24.8	21.0	18.8		
	10%	WB	13.7	14.7	16.4	18.9	21.8	24.0	24.9	24.7	23.0	20.3	16.4	14.7		
		MCDB	16.0	17.2	20.3	23.1	26.6	29.2	30.1	29.8	27.3	23.5	19.3	16.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.3	11.7	12.5	13.1	12.0	11.0	10.6	10.7	11.4	12.6	12.8	11.0		
		MCDBR	13.1	13.6	14.9	14.9	13.5	13.2	12.3	12.9	13.6	13.9	14.1	12.4		
	5% WB	MCWBR	9.0	8.6	7.6	6.8	5.5	4.8	4.2	4.1	4.8	6.2	8.1	8.7		
		MCWBR	10.8	11.2	11.8	11.4	11.3	10.9	10.6	10.7	10.2	9.9	10.6	10.6		
(40) Clear-Sky Solar Irradiance	taub		0.321	0.325	0.362	0.394	0.469	0.488	0.537	0.538	0.433	0.374	0.344	0.324		
	taud		2.508	2.494	2.413	2.307	2.136	2.131	2.008	1.993	2.278	2.430	2.453	2.515		
	Ebn at Noon		894	928	914	894	827	808	766	758	830	860	856	869		
	Edh at Noon		85	95	111	129	155	156	175	175	126	100	89	80		
(44) All-Sky Solar Radiation	RadAvg		2.64	3.17	4.29	5.50	6.01	6.12	5.91	5.46	4.76	4.01	2.97	2.25		
	RadStd		0.18	0.41	0.39	0.37	0.50	0.48	0.41	0.31	0.44	0.52	0.32	0.23		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+101	+70
(47) Regional (47 neighbors)	+0.49	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+159	+113

Nomenclature: See separate page